The family of Gaussian mixture model (GMM) with prescribed component distributions (ie., convex weight combinations of $D+1$ Gaussian densities) form a mixture family [93].

Notice that the probability simplex of discrete distributions can be *both* modeled as an exponential family or a mixture family [8].

The *expected α-geometry* is built from the *expected dual $\pm\alpha$-connections*. The Fisher "*information metric*" tensor is built from the FIM as follows:

$$ {}_{\mathcal{P}}g(u,v) := (u)_\theta^\top \, {}_{\mathcal{P}}I(\theta) \, (v)_\theta \tag{95} $$

The expected *exponential connection* and expected *mixture connection* are given by

$$ {}^{e}_{\mathcal{P}}\nabla := E_\theta \left[(\partial_i \partial_j l)(\partial_k l) \right], \tag{96} $$

$$ {}^{m}_{\mathcal{P}}\nabla := E_\theta \left[(\partial_i \partial_j l + \partial_i l \partial_j l)(\partial_k l) \right]. \tag{97} $$

The dualistic structure is denoted by $(\mathcal{P}, {}_{\mathcal{P}}g, {}^{m}_{\mathcal{P}}\nabla, {}^{e}_{\mathcal{P}}\nabla)$ with Amari-Chentsov cubic tensor called the *skewness tensor*:

$$ C_{ijk} := E_\theta \left[\partial_i l \partial_j l \partial_k l \right]. \tag{98} $$

It follows that we can build a one-family of expected information α-manifolds:

$$ \left\{ (\mathcal{P}, {}_{\mathcal{P}}g, {}_{\mathcal{P}}\nabla^{-\alpha}, {}_{\mathcal{P}}\nabla^{+\alpha}) \right\}_{\alpha \in \mathbb{R}}, \tag{99} $$

with

$$ {}_{\mathcal{P}}\Gamma^{\alpha}{}_{ij,k}(\theta) := E_\theta \left[\partial_i \partial_j l \partial_k l \right] + \frac{1-\alpha}{2} C_{ijk}(\theta), \tag{100} $$

$$ = E_\theta \left[\left(\partial_i \partial_j l + \frac{1-\alpha}{2} \partial_i l \partial_j l \right) (\partial_k l) \right]. \tag{101} $$

The Levi-Civita metric connection is recovered as follows:

$$ {}_{\mathcal{P}}\bar{\nabla} = \frac{{}_{\mathcal{P}}\nabla^{-\alpha} + {}_{\mathcal{P}}\nabla^{\alpha}}{2} = {}^{\mathrm{LC}}_{\mathcal{P}}\nabla := {}^{\mathrm{LC}}\nabla({}_{\mathcal{P}}g) \tag{102} $$

The α-Riemann-Christoffel curvature tensor is:

$$ {}_{\mathcal{P}}R_{ijkl} = \partial_i \Gamma^{\alpha}_{jk,l} - \partial_j \Gamma^{\alpha}_{ik,l} + g^{rs} \left(\Gamma^{\alpha}_{ik,r} \Gamma^{\alpha}_{js,l} - \Gamma^{\alpha}_{jk,r} \Gamma^{\alpha}_{is,l} \right), \tag{103} $$

with $R^{\alpha}_{ijkl} = -R^{-\alpha}_{ijlk}$. We check that the expected $\pm\alpha$-connections are coupled with the metric: $\partial_i g_{jk} = \Gamma^{\alpha}_{ij,k} + \Gamma^{-\alpha}_{ik,j}$.

In case of an exponential family $\mathcal{E}$ or a mixture family $\mathcal{M}$ equipped with the dual exponential/mixture connection, we get *dually flat manifolds* (Bregman geometry).

Indeed, for the exponential/mixture families, it is easy to check that the Christoffel symbols of ∇^e and ∇^m vanish:

$$ {}^{e}_{\mathcal{M}}\Gamma = {}^{m}_{\mathcal{M}}\Gamma = {}^{e}_{\mathcal{E}}\Gamma = {}^{m}_{\mathcal{E}}\Gamma = 0. \tag{104} $$

3.10 Criteria for statistical invariance

So far we have explained how to build an information manifold (or information α-manifold) from a pair of conjugate connections. Then we reported two ways to obtain such a pair of conjugate connections: (1) from a parametric divergence, or (2) by using the predefined expected exponential/mixture connections. We now ask the following question: Which information manifold makes sense in Statistics? We can refine the question as follows:

- Which metric tensors g make sense in statistics?

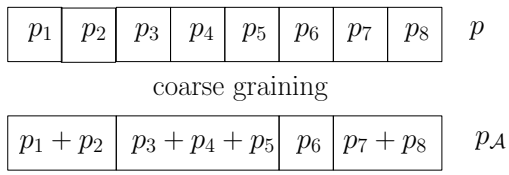

Figure 9: A divergence satisfies the property of information monotonicity iff $D(\theta_{\bar{\mathcal{A}}} : \theta'_{\bar{\mathcal{A}}}) \leq D(\theta : \theta')$. Here, parameter θ represents a discrete distribution.

- Which affine connections ∇ make sense in statistics?
- Which statistical divergences make sense in statistics (from which we can get the metric tensor and dual connections)?

By definition, an *invariant metric tensor* g shall preserve the inner product under important *statistical mappings* called Markov embeddings. Informally, we embed Δ_D into $\Delta_{D'}$ with $D' > D$ and the induced metric should be preserved (see [8], page 62).

Theorem 8 (Uniqueness of Fisher information metric [26, 129])**.** *The Fisher information metric is the unique invariant metric tensor under Markov embeddings up to a scaling constant.*

A D-dimensional parameter (discrete) divergence satisfies the *information monotonicity*[12] if and only if:

$$D(\theta_{\bar{\mathcal{A}}} : \theta'_{\bar{\mathcal{A}}}) \leq D(\theta : \theta') \tag{105}$$

for any *coarse-grained partition* $\mathcal{A} = \{\mathcal{A}_i\}_{i=1}^E$ of $[D] = \{1, \ldots, D\}$ ($\mathcal{A}$-lumping [34]) with $E \leq D$, where $\theta^i_{\bar{\mathcal{A}}} = \sum_{j \in \mathcal{A}_i} \theta^j$ for $i \in [E]$. This concept of coarse-graining is illustrated in Figure 9.

A *separable divergence* $D(\theta_1 : \theta_2)$ is a divergence that can be expressed as the sum of elementary *scalar divergences* $d(x : y)$:

$$D(\theta_1 : \theta_2) := \sum_i d(\theta_1^i : \theta_2^j). \tag{106}$$

For example, the squared Euclidean distance $D(\theta_1 : \theta_2) = \sum_i (\theta_1^i - \theta_2^i)^2$ is a separable divergence for the scalar Euclidean divergence $d(x : y) = (x - y)^2$. The Euclidean distance $D_E(\theta_1, \theta_2) = \sqrt{\sum_i (\theta_1^i - \theta_2^i)^2}$ is *not* separable because of the square root operation.

The only invariant and *decomposable* divergences when $D > 1$ are f-divergences [56] defined for a convex functional generator f:

$$I_f(\theta : \theta') := \sum_{i=1}^D \theta_i f\left(\frac{\theta'_i}{\theta_i}\right) \geq f(1), \quad f(1) = 0 \tag{107}$$

The *standard f-divergences* are defined for f-generators satisfying $f'(1) = 0$ (choose $f_\lambda(u) := f(u) + \lambda(u-1)$ since $I_{f_\lambda} = I_f$), and $f''(u) = 1$ (scale fixed).

Statistical f-divergences are *invariant* [108] under one-to-one/sufficient statistic transformations $y = t(x)$ of sample space: $p(x; \theta) = q(y(x); \theta)$:

$$\begin{aligned} I_f[p(x;\theta) : p(x;\theta')] &= \int_{\mathcal{X}} p(x;\theta) f\left(\frac{p(x;\theta')}{p(x;\theta)}\right) \mathrm{d}\mu(x), \\ &= \int_{\mathcal{Y}} q(y;\theta) f\left(\frac{q(y;\theta')}{q(y;\theta)}\right) \mathrm{d}\mu(y), \\ &= I_f[q(y;\theta) : q(y;\theta')]. \end{aligned}$$

[12] This property could be renamed as the "distance coarse-binning inequality property."

The dual f-divergences for reference duality is

$$I_f{}^*[p(x;\theta):p(x;\theta')] = I_f[p(x;\theta'):p(x;\theta)] = I_{f^\diamond}[p(x;\theta):p(x;\theta')] \tag{108}$$

for the standard conjugate f-generator (diamond $f^\diamond$ generator) with:

$$f^\diamond(u) := uf\left(\frac{1}{u}\right). \tag{109}$$

One can check that $f^\diamond$ is a standard f-generator when f is standard.

Let us report some common examples of f-divergences:

- The family of α-*divergences*:

$$I_\alpha[p:q] := \frac{4}{1-\alpha^2}\left(1-\int p^{\frac{1-\alpha}{2}}(x)q^{\frac{1+\alpha}{2}}(x)\mathrm{d}\mu(x)\right), \tag{110}$$

  obtained for $f(u) = \frac{4}{1-\alpha^2}(1-u^{\frac{1+\alpha}{2}})$. The α-divergences include:

  - the *Kullback-Leibler* when $\alpha \to 1$:

$$\mathrm{KL}[p:q] = \int p(x)\log\frac{p(x)}{q(x)}\mathrm{d}\mu(x), \tag{111}$$

    for $f(u) = -\log u$.
  - the *reverse Kullback-Leibler* $\alpha \to -1$:

$$\mathrm{KL}^*[p:q] = \int q(x)\log\frac{q(x)}{p(x)}\mathrm{d}\mu(x) = \mathrm{KL}[q:p], \tag{112}$$

    for $f(u) = u\log u$.
  - the symmetric squared *Hellinger divergence*:

$$H^2[p:q] = \int (\sqrt{p(x)}-\sqrt{q(x)})^2\mathrm{d}\mu(x), \tag{113}$$

    for $f(u) = (\sqrt{u}-1)^2$ (corresponding to $\alpha = 0$)
  - the Pearson and Neyman chi-squared divergences [90], etc.

- the *Jensen-Shannon divergence*:

$$\mathrm{JS}[p:q] = \frac{1}{2}\int\left(p(x)\log\frac{2p(x)}{p(x)+q(x)} + q(x)\log\frac{2q(x)}{p(x)+q(x)}\right)\mathrm{d}\mu(x), \tag{114}$$

  for $f(u) = -(u+1)\log\frac{1+u}{2} + u\log u$.

- the *Total Variation*

$$\mathrm{TV}[p:q] = \frac{1}{2}\int |p(x)-q(x)|\mathrm{d}\mu(x), \tag{115}$$

  for $f(u) = \frac{1}{2}|u-1|$. The total variation distance is the only metric f-divergence.

The f-topology is the topology generated by open f-balls, open balls with respect to f-divergences. Atopology$T$ is saidstrongerthan atopologyT' if T contains all the open sets of T'. Csiszar's theorem [33] states that when $|\alpha| < 1$, the α-topology is equivalent to the topology induced by the total variation metric distance. Otherwise, the α-topology is stronger than the TV topology.

Let us state an important feature of f divergences: